

peak demand and more than 24 per cent T&D losses. In most rural areas, electricity is not available for more than eight hours a day, so it is necessary to move towards energy-efficient techniques with renewable energy support.

Low-voltage DC distribution system topologies

A low-voltage DC (LVDC) system is of two types, according to conductor polarities:

Unipolar LVDC system. In this type of distribution system, voltage level throughout the system is the same. The two-conductor system has one conductor at positive polarity and another at zero or neutral polarity. Entire load on the distribution system is connected at this voltage level, as shown in Fig. 1.

Bipolar LVDC system. A bipolar LVDC system is a combination of two series-connected unipolar systems. In this system, it is the choice of the customer to connect to the appropriate voltage level. The choices are:

1. Positive to neutral polarity
2. Negative to neutral polarity
3. Positive to negative polarity
4. Positive to negative polarity with neutral connection

Fig. 2 shows the possible alternatives for a consumer connection. Customer connections 1 and 2 can lead to an unsymmetrical loading situation between DC poles in the system. Possible over-voltage can be restricted with a cable cross-

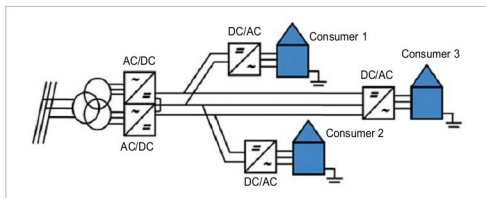


Fig. 2: Bipolar LVDC distribution system with different customer connection alternatives

section selection. Connections 1 and 2 are used in studied $\pm 750V$ DC bipolar system. Main lines of the system contain all three conductors but customer connections have two-wire cables connected between positive or negative poles. Therefore customer supply voltage is either $+750V$ DC or $-750V$ DC.

Energy efficiency analysis of AC/DC, DC/DC and DC/AC conversions

According to the survey of different products in the market, some

conversion efficiencies for different wattage ratings are given in Table I, which can be further used for calculations.

Losses calculation

Loads in the building. Academic buildings consist of different connected loads, broadly mentioned in Table II with their wattage ratings. From Table I it is clear that, according to the wattage rating, conversion efficiency of every electrical equipment is different. Now, to calculate the total energy consumed by each equipment, a site survey was done. According to the site survey, total electricity consumed by a particular load is given in Table II.

Power supplies of loads in the building. From the above-mentioned data of connected loads for the buildings, conversion efficiency formulae for different loads are given as follows:

Desktop computers. Efficiency for desktop computers on existing AC supply is given by:

$$\eta_{AC} = \eta_{UPS} \times \eta_{ACDC} \times \eta_{DCDC} \times \eta_{Feeder} \quad \dots (1)$$

If the same is used to supply from DC power system, power efficiency is:

$$\eta_{DC} = \eta_{UPS} \times \eta_{DCDC} \times \eta_{Feeder} \quad \dots (2)$$

where η_{UPS} is the power efficiency of UPS (75 per cent taken here by measurement), η_{ACDC} is the power efficiency of transformer, η_{DCDC} is the power efficiency of AC/DC, η_{DCDC} is the power efficiency of DC/DC and η_{Feeder} is the power efficiency of feeder, which is 100 per cent.

TABLE I
CONVERSION EFFICIENCIES OF DIFFERENT ELECTRICAL EQUIPMENT BY WATTAGE

Power rating (W)	10	100	1000	10,000	100,000	1,000,000
The conversion						
AC/DC	69%	75%	81%	87%	93%	99%
DC/DC	86%	90%	92%	96%	97%	99%
DC/AC	90%	92%	92%	95%	95%	98%

TABLE II
STATISTICS OF LOAD IN THE ACADEMIC BUILDING OF MCOE AND RC NASHIK

Loads	Power (W)	Quantity	Total (W)
Fluorescent lamps	40	419	16,760
Fans	80	842	67,360
Desktop computers	150	548	82,200
Printers	40	48	1920
IT devices	20	27	540
Photocopy machines	200	04	800
Water coolers	1000	08	8000
Water pumps	2332	02	4664
LED lights	68	20	1360